

Privilege Escalation and Arbitrary Page Tampering in Cloudflare Pages

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Content (original)

The source's own words. An English translation of this document is archived beside it as `2023-blog-ryotak-net-cloudflare-pages_translate.md`.

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Cloudflare Pages

* 2023-12-23 * 5710 **Cloudflare Python

You can read about these vulnerabilities in English at <https://ec0.io/post/hacking-cloudflare-pages-part-2/>

Cloudflare	HackerOne	(Bug Bounty)
	Cloudflare	
	Cloudflare	
Cloudflare		Cloudflare
	2022	

Cloudflare Pages

Cloudflare Pages Cloudflare JAMstack
Cloudflare Workers

Cloudflare Pages, part 1: The fellowship of the secret
Cloudflare Pages

Cloudflare Pages
Kubernetes
1

buildbot

USER	PID	%CPU	MEM	VSZ	RES	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.2	53196	32788	?	Ss	08:03	0:00	/bin/bash /opt/root/gke-build.sh
root	4	0.0	0.1	53196	31032	?	S	08:03	0:00	/bin/bash /opt/pages/build_tool/run_build.sh
root	5	0.1	0.5	4648248	84576	?	Sl	08:03	0:00	_ /opt/pages/bin/live_logs --log-file /opt/pages/user_logs.log --junit-file /opt/pages/config/JWT --host-file /opt/pages/config/MAESTRO_HOST --f
features-file	/opt/pages/config/FEATURES									
root	6	0.7	0.6	909948	105828	?	Sl	08:03	0:00	_ node heartbeat.js
root	9	0.0	0.1	53200	30788	?	S	08:03	0:00	_ /bin/bash /opt/pages/build_tool/run_build.sh
root	10	0.0	0.1	39508	29712	?	S	08:03	0:00	_ tee -a /opt/pages/build_tool.log
root	7	1.0	0.3	110620	54460	?	S	08:03	0:00	_ /usr/bin/python3.7 -u /opt/pages/build_tool/main.py
root	82	0.0	0.2	82648	33876	?	S	08:03	0:00	_ sudo PATH=/opt/buildhome/.swiftenv/bin:/opt/buildhome/.swiftenv/shims:/opt/buildhome/.php:/opt/buildhome/.binrc/bin:/usr/local/rvm/bin:/
usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/opt/buildhome/.cask/bin:/opt/buildhome/.gimme/bin:/opt/buildhome/.dotnet/tools:/opt/buildhome/.dotnet -H -E -u buildbot /opt/build/bin/build bash -i										
buildbot	83	4.4	0.2	51956	38832	?	S	08:03	0:01	_ /bin/bash /opt/build/bin/build bash -i
buildbot	1287	0.1	0.2	53376	31984	?	S	08:03	0:00	_ bash -i
buildbot	1292	0.0	0.2	69500	34236	?	R	08:03	0:00	_ ps auxf
root	8	0.0	0.1	39508	29952	?	S	08:03	0:00	_ tee -a /opt/pages/build_tool.log

/opt/pages/build_tool/main.py (build_tool)

sudo

build_tool

root

build_tool

build_tool

build_tool

Cloudflare

root

root

index.html

/etc/shadow

index.html

Cloudflare Pages

Redirects

_redirects

Redirects

Creating redirects

To use redirects on Cloudflare Pages, declare your redirects in a plain text file called `_redirects` without a file extension, in the output folder of your project. The `build output folder` is project-specific, so the `_redirects` file should not always be in the root directory of the repository. Changes to redirects will be updated to your website at build time so make sure you commit and push the file to trigger a new build each time you update redirects.

: [Cloudflare Docs](#)

buildbot

/etc/shadow

Redirects

Redirects configured for this deployment. [Creating redirects](#)

```
root:*:18432:0:99999:7:::
daemon:*:18432:0:99999:7:::
bin:*:18432:0:99999:7:::
sys:*:18432:0:99999:7:::
sync:*:18432:0:99999:7:::
games:*:18432:0:99999:7:::
man:*:18432:0:99999:7:::
lp:*:18432:0:99999:7:::
mail:*:18432:0:99999:7:::
news:*:18432:0:99999:7:::
uucp:*:18432:0:99999:7:::
proxy:*:18432:0:99999:7:::
www-data:*:18432:0:99999:7:::
```

21

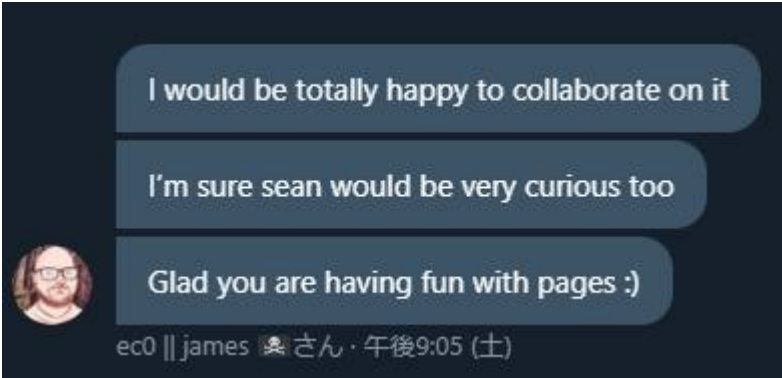
1

2

2

[[image: Twitter](#) 2]

James 3



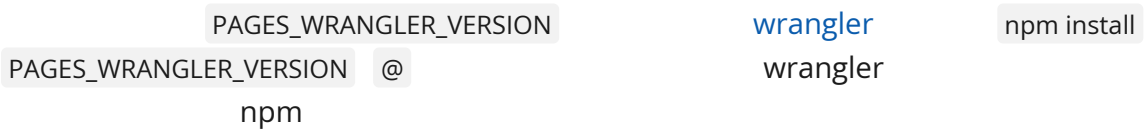
build_tool



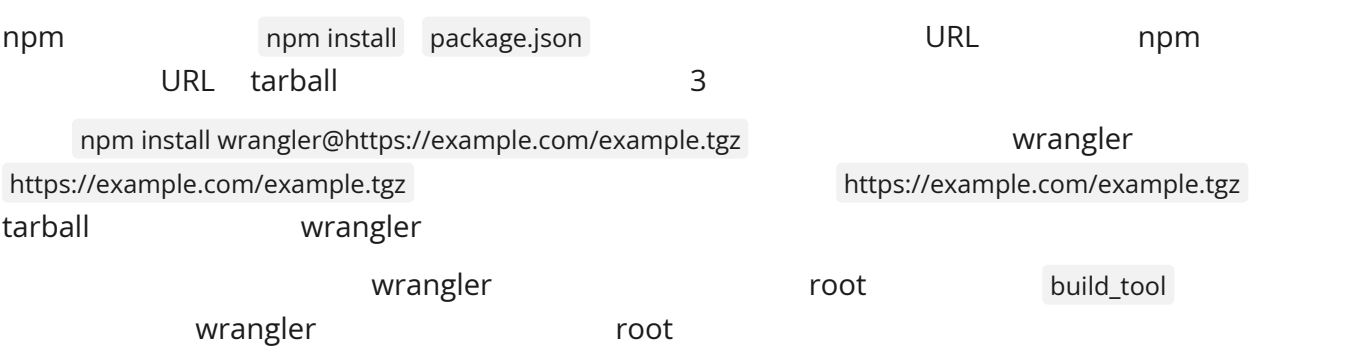
```
version = [env_var['value'] for env_var in env_vars if env_var['key'] == 'PAGES_WRANGLER_VERSION'][0]
print_line(f'Overriding wrangler version to {version}...', logs)

subprocess.run(['npm', 'install', f'wrangler@{version}'], cwd=WRANGLER_DIR, check=True, capture_output=True)

print_line('wrangler version override complete!', logs)
```



npm



```
cmd = [  
    './node_modules/.bin/wrangler2',  
    'pages',  
    'functions',  
    'build',  
    "--outfile",  
    constants.OUTPUT_WORKER_PATH,  
    "--output-config-path",  
    constants.USER_WORKER_DERIVED_CONFIG_PATH,  
    '--build-output-directory',  
    output_dir,  
    functions_dir  
]
```

with subprocess.Popen(cmd, **plinko_args) as proc:

root

[image: Discord npm]

root

Sean 3

Cloudflare

build_tool

wrkr

Rust

build_tool

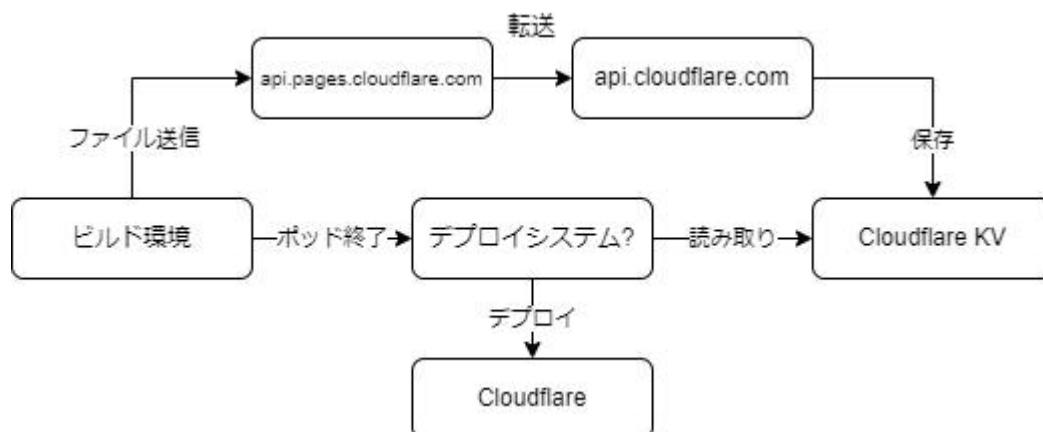
```
# Configure api proxy through the maestro
os.environ['CF_API_HOST'] = config['MAESTRO_HOST']
os.environ['CF_API_TOKEN'] = config['JWT']
os.environ['CF_ACCOUNT_ID'] = config['CF_ACCOUNT_ID']
[...]

# WRKR (ASSET UPLOADER)
upload_args = {
    'account_tag': config['CF_ACCOUNT_ID'],
    'asset_namespace': config['ASSET_NAMESPACE'],
    'asset_dir': asset_dir,
    'asset_manifest_namespace': config['MANIFEST_NAMESPACE'],
    'asset_manifest_key': config['MANIFEST_KEY'],
}

cmd = ['./wrkr']

[...]
lines = filter_logs(util.run_cmd(cmd, cwd=constants.WRKR_DIR))
```

wrkr CF_API_HOST
api.pages.cloudflare.com



API

api.pages.cloudflare.com API

:

- /client/v4/accounts/d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/db62b722715546c9af0cedbd574c9a47/bulk
- /client/v4/accounts/d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/db62b722715546c9af0cedbd574c9a47/keys

- ```
/client/v4/accounts/d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/332a39fcd8a845d7909d2d5d753604d8/values/builds/5486590/logs
```

|     |                    |
|-----|--------------------|
| API | api.cloudflare.com |
| JWT | api.cloudflare.com |

API

API

API

3

ID( d6fa5e8917ff81a61c1f92fc98b9f85d )

 **ec0** 2022/05/15  
d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/db62b722715546c9af0cedbd574c9a47  
these parameters seem the same on my build let me closely check

 **minight** 2022/05/15  
d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/332a39fcd8a845d7909d2d5d753604d8  
d6fa5e8917ff81a61c1f92fc98b9f85d/storage/kv/namespaces/db62b722715546c9af0cedbd574c9a47  
it might be  
that its a different namespace

|    |   |   |
|----|---|---|
| KV | 2 | 1 |
|----|---|---|

JWT

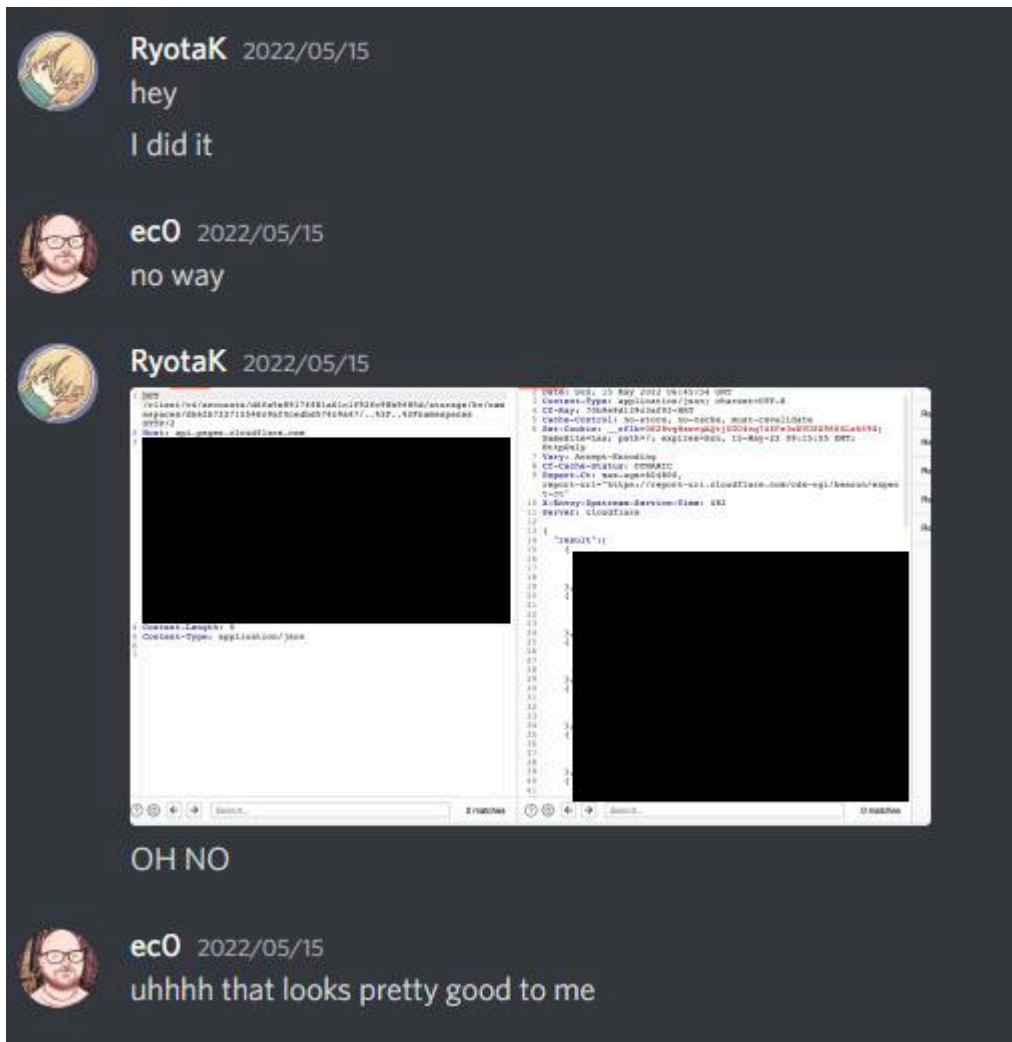
ID

Sean

API

[image: Sean    Discord    ]

API



..%2F..%2F..%2F

api.cloudflare.com

API

API

API

Pages

API

Pages

Cloudflare Workers KV

Cloudflare Workers

Cloudflare Pages

Cloudflare Pages

/ Twitter(@ryotkak)



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|            |                    |                  |            |
|------------|--------------------|------------------|------------|
| ( )        | 2022/05/14         | 2022/05/14       |            |
| 2022/05/15 | 2022/05/15   Pages | API              | 2022/05/15 |
| Cloudflare | 3                  | 2022/05/17   API | 2022/06/07 |
|            | 2022/06/08         |                  | 2023/12/22 |
| 2023/12/23 |                    |                  |            |

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<https://github.com/npm/npm-package-arg/blob/2dd33f52a772c091f26169c97cefaa399a7233cc/lib/npa.js#L77-L85>

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